Big Ben Logistics

Dispatch Scheduler System Manual

ISO 9001:2015 Quality Management System

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Document Control

This page records the revision history of, and the formal approvals required for, the Dispatch Scheduler System Manual. It satisfies the requirements of ISO 9001:2015 Clause 7.5.3 (Control of documented information).

Revision History

Rev.	Date	Section(s)	Description of Change	Author
1.0	May 2026	All	Initial release.	IT / Operations
1.1	May 2026	B.1, B.2, B.7, C-004, E.3, F.3	Truck Cycle Time redesigned to plate-centric view (Plate Status + cycle picker + audit trail timeline). Toll auto-fill decoupled from Schedule — manual entry remains the source of truth for billing; GPS-detected tolls surface as a separate Dashboard KPI and on the Toll Log page. New Schedule cross-wave copy dropdown. New Settings modal on TCT for configurable dwell + stop-detection thresholds. New admin-only Clear Logs with type-CLEAR confirmation. New manual geofences workaround for any future API visibility issues. Ad-hoc stop detection (≥ N min outside any geofence). Timezone serialisation unified to PHT-aware ISO.	IT / Operations

Approval

By signing below, the listed officers confirm that this manual accurately describes the current operational use of the Dispatch Scheduler System and authorise its release as a controlled document.

Role	Name	Signature	Date
Prepared by	[TO BE FILLED]		
Reviewed by	[TO BE FILLED]		
Approved by	[TO BE FILLED]		

Role	Name	Signature	Date
Mgmt. Rep. (ISO)	[TO BE FILLED]		

Distribution List

This manual is distributed to the following roles. Each holder is responsible for maintaining the integrity of their copy and destroying obsolete versions when superseded.

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01	Operations Manager	Operations	Digital (PDF)
02	Dispatch Supervisor	Operations	Digital (PDF) + Print
03	Fleet Manager	Fleet	Digital (PDF)
04	IT Administrator	IT / Systems	Digital (PDF)
05	HR / Training	Human Resources	Digital (PDF)

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Section page numbers are generated automatically from the document structure.

Glossary & Acronyms

The following terms are used throughout this manual.

Term / Acronym	Definition
API	Application Programming Interface — the protocol used by external systems (Cartrack, Google Sheets) to exchange data with the Dispatch Scheduler.
Auto-fill	The automatic populating of a TripRecord field (such as toll fee) from GPS telemetry, without manual data entry.
BIG BEN SCM	Big Ben Supply Chain Management — the home base geofence. Cycles open when a truck leaves this geofence and close when it returns.
Breakdown	A recorded incident in which a vehicle becomes unavailable due to mechanical, electrical, or accident-related causes.
Cartrack	The fleet GPS tracking provider integrated with this system. Provides real-time vehicle positions, geofence events, and trip data.
Cycle	One complete round trip: a truck exits the home geofence, performs deliveries or pickups, and returns to the home geofence.
Dispatcher	The operations staff member responsible for assigning trips to drivers and tracking their status.
Drive-by	A geofence visit shorter than the minimum dwell time (default 5 minutes) — treated as a transient touch, not a delivery stop.
Geofence	A virtual boundary drawn around a real-world location (customer site, quarry, toll plaza, fuel station) used to detect vehicle entry and exit.
Idling	A state in which the engine is running but the vehicle is stationary, as reported by the Cartrack telemetry.
ISO 9001	International standard for Quality Management Systems (current revision: ISO 9001:2015).
KPI	Key Performance Indicator. Examples in this system: trips per day, fleet utilisation, average cycle time, total toll spend.
Master Data	The reference data shared across the application: drivers, helpers, plates, products, clients, dispatchers, truck types.
Plate	A vehicle in the fleet, identified by its registration (plate number) and an internal body number (e.g., DT15, TH08).
Plaza	A toll collection point on a Philippine expressway. The system maintains a matrix of inter-plaza toll fees for auto-calculation.

Term / Acronym	Definition
PHT	Philippine Time (UTC+8) — the time zone used for all business dates and schedules.
SOP	Standard Operating Procedure — a documented sequence of steps for executing a recurring task.
TripRecord	The database record representing one delivery or pickup assigned to a driver, helper, and plate within a Wave.
Truck Type	A classification of vehicles by capacity and use (e.g., DT — Dump Truck, TH — Trailer Hauler, MDT — Mini Dump Truck).
UTC	Coordinated Universal Time. The server runs in UTC; all displayed times are converted to PHT.
Wave	A scheduled batch of trips for a given date and truck type. A day can have multiple waves (e.g., morning and afternoon).
WSGI	Web Server Gateway Interface — the Python web-app deployment standard used on PythonAnywhere.

Section A — System Overview

This section introduces the Dispatch Scheduler System: its purpose, business context, architecture, modules, and the user roles that interact with it. It provides the foundation for the detailed user procedures and Standard Operating Procedures that follow in subsequent sections.

A.1 Purpose and Scope

The Dispatch Scheduler is the central operating system for managing day-to-day fleet operations at Big Ben Logistics. It replaces the previous combination of paper-based dispatch sheets, Excel logs, and ad-hoc messaging applications with a single integrated platform accessible from any web browser.

The system covers the full operational cycle of a delivery or pickup: from the moment a trip is scheduled, through the assignment of driver, helper, and plate, the live tracking of the truck via GPS, the recording of toll fees and breakdowns, and finally the closing of the trip with all supporting documentation.

ISO 9001 REFERENCE Scope of this Manual

This manual describes the operational use of the Dispatch Scheduler Application as deployed on the Big Ben Logistics PythonAnywhere environment. It covers all modules visible to authenticated users. Source-code development, infrastructure administration, and external vendor (Cartrack, Google Sheets) configuration are outside its scope.

A.2 Business Context

Big Ben Logistics operates a mixed fleet of approximately fifty trucks across multiple categories — Dump Trucks (DT), Trailer Haulers (TH), Mini Dump Trucks (MDT), Self-Loading Trucks, and utility vehicles such as L300 vans. The fleet services construction projects, quarry-to-batching-plant routes, and general logistics across Luzon.

A typical operating day begins before sunrise, with multiple trips per truck and frequent expressway transits. The system is designed to handle the operational complexity that arises from this scale: dozens of concurrent trips, multi-day journeys, shared drivers across truck types, breakdown rotations, and the corresponding toll-fee, fuel, and driver-incentive calculations.

Key Operational Inputs

- Trip plans confirmed the previous evening or early morning.
- Driver and helper attendance, recorded daily.
- Plate availability — accounting for breakdowns and maintenance.
- Customer purchase orders, delivery receipts, and reference numbers.
- Toll-plaza coordinates and inter-plaza fee matrices.
- Real-time GPS telemetry from Cartrack (vehicle position, speed, geofence events).

Key Operational Outputs

- A full audit trail of every assigned, completed, or cancelled trip.
- Daily, weekly, and monthly key-performance indicators (KPIs).
- Auto-filled toll fees per trip, where GPS evidence is available.
- Cycle-time analytics — how long each truck spends per round trip.
- Idling-rate analytics at customer and quarry geofences.
- Exportable reports in Excel and Google Sheets formats for finance, payroll, and management review.

A.3 System Architecture

The Dispatch Scheduler follows a three-tier web-application architecture. The presentation tier renders to any modern browser. The application tier runs a Flask Python application hosted on PythonAnywhere. The data tier is a SQLite database that persists all operational records. Real-time GPS data is consumed from Cartrack via a REST API and stored in the same database for unified reporting.

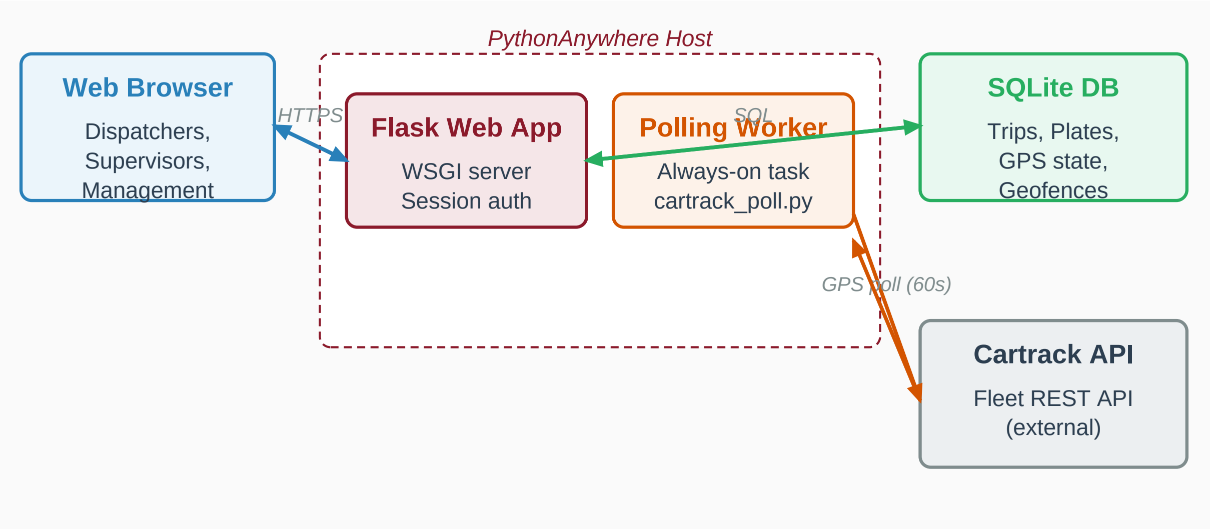

Figure A-1: High-level system architecture. Browser sessions reach the Flask web app over HTTPS; a parallel always-on polling worker ingests GPS data from the Cartrack API. Both processes share the same SQLite database.

Major Components

Component	Technology	Function
Web UI	HTML5, Bootstrap 5, vanilla JS, Chart.js	Browser-rendered pages for dispatch staff, supervisors, and management.
Application	Python 3, Flask, Flask-Login, SQLAlchemy	Routes requests, validates input, applies business rules, returns JSON or HTML.
Database	SQLite 3	Stores trips, waves, master data, breakdowns, GPS state, geofences, cycles.
GPS provider	Cartrack Fleet REST API	Vehicle positions, geofence events, ignition / idling status, trip history.
Sheets sync	Google Apps Script webhook	Outbound push of structured data to a shared Google Sheet for finance.
Polling worker	Standalone Python script (cartrack_poll.py)	Runs continuously on a PythonAnywhere always-on task to ingest GPS data.
Authentication	Username + password, hashed (Flask-Login session)	Restricts every page to authenticated users.

Hosting & Availability

The production environment is hosted on PythonAnywhere under a paid Developer tier. The application is reachable at a https URL and is restricted to authenticated Big Ben Logistics users. The polling worker runs as an always-on task on the same host, so GPS data continues to flow even when no operator is logged into the web interface. Daily CPU budget is approximately five thousand seconds, which comfortably accommodates a sixty-second polling cadence across the full fleet.

A.4 Modules Overview

The application is organised into nine functional modules, each accessible from the left-hand sidebar.

#	Module	Primary Users	Purpose
1	Dashboard	All	Snapshot of today's operations — trip counts, fleet utilisation, cycle time, breakdowns, KPIs.
2	Schedule	Dispatchers, Supervisors	Create and edit the daily trip plan, assign personnel and plates, update trip status.
3	Master Data	Operations Manager, IT	Manage drivers, helpers, plates, products, clients, dispatchers, truck types, and GPS mappings.
4	Breakdown	Fleet Manager, Mechanics	Record incidents that take a plate out of service; track repair status and downtime.
5	Toll Calculator	Dispatchers	Compute the expected toll fee between any two plazas on any expressway, with multi-route support.
6	Toll Log	Dispatchers, Finance	View every GPS-detected plaza event in chronological order; export to Excel.

#	Module	Primary Users	Purpose
7	Truck Cycle Time	Fleet Manager, Management	Live plate status, per-truck audit trail (visits, plaza events, in-progress stops), idling-rate ranking, multi-day cycle tracking. Configurable thresholds and admin Clear Logs.
8	Reports	Management, Finance	Aggregated KPIs and trend charts across configurable date ranges.
9	Admin / Settings	IT, Management Representative	User account management, password resets, environment configuration.

A.5 User Roles and Responsibilities

The system uses session-based authentication. There are no in-application permission tiers at the time of writing — every authenticated user has access to every module. Access control is therefore exercised through account provisioning: only the roles listed below are granted accounts, and each user is expected to operate within the responsibilities described.

Role	Modules Used (Primary)	Key Responsibilities
Dispatcher	Schedule, Toll Calculator, Toll Log	Build the daily plan, assign trips, update statuses, verify auto-filled tolls.
Dispatch Supervisor	Schedule, Dashboard, Reports	Approve the plan, monitor execution, intervene on exceptions, sign off on daily totals.
Fleet Manager	Breakdown, Truck Cycle Time, Master Data	Maintain plate availability, investigate long cycle times, manage maintenance schedules.
Operations Manager	All	Cross-functional oversight; final accountability for operational metrics.
Finance Officer	Toll Log, Reports	Reconcile auto-filled tolls against receipts; produce monthly cost reports.
IT Administrator	Admin / Settings, Master Data	Provision and de-provision user accounts; manage environment variables and integrations.
Management Representative (ISO)	All	Custodian of this manual; conducts periodic reviews and authorises revisions.

> **ISO 9001 REFERENCE Account Provisioning**
>
> New user accounts are created exclusively by the IT Administrator on written request from the Operations Manager. Accounts of staff leaving the organisation are deactivated within one business day of separation. See SOP-005 (Data Backup and User Access Management).

A.6 Operating Environment

Client-Side Requirements

• Any modern web browser released within the last three years — Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari.

• A stable internet connection. Mobile data is sufficient for occasional checks but a wired or Wi-Fi connection is recommended for dispatch workstations.

• A screen resolution of at least 1280 by 720 pixels. The interface adapts to smaller devices but is optimised for a full-size workstation monitor.

Time Zone Convention

All business dates and schedules use Philippine Time (UTC+8). The application server runs in UTC internally and converts to PHT at display time. Operators are not expected to perform time-zone arithmetic.

Network Dependencies

The application requires outbound HTTPS access to the Cartrack Fleet API endpoint and, when configured, to the Google Apps Script webhook. Both endpoints must be reachable from the PythonAnywhere host. The IT Administrator maintains the allow-list of permitted external services.

Section B — User Manual

This section is a step-by-step operating guide for each of the nine modules in the Dispatch Scheduler. Each subsection follows the same template: an introduction, a screen-by-screen walk through of the user interface, the most common tasks performed in the module, and any quality-control checks the user must perform before considering a task complete.

B.0 Login and Session Management

Every page of the application requires authentication. Visiting any URL without an active session redirects the browser to the login screen.

Logging In

1. Open a web browser and navigate to the application URL provided by the IT Administrator.

2. Enter your assigned username in the **Username** field.

3. Enter your password in the **Password** field. Passwords are case-sensitive.

4. Click **Sign In**. On success you will be redirected to the Dashboard.

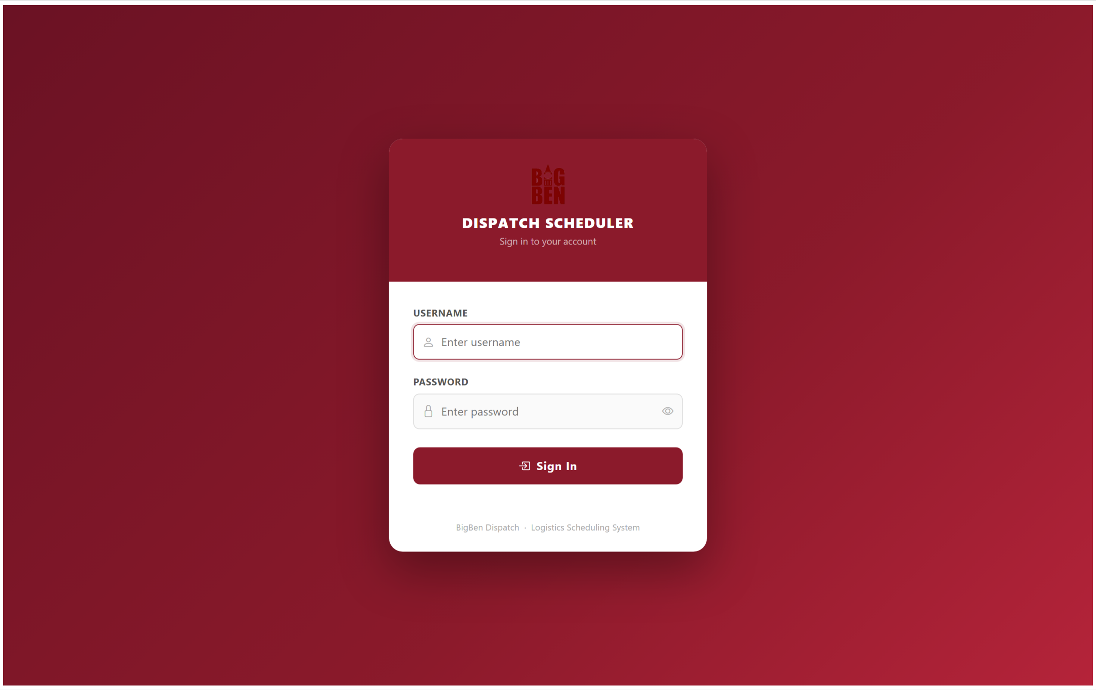

Figure: Login screen

Logging Out

Click your username in the top-right corner of any page, then choose **Sign Out**. Always sign out when leaving a shared workstation unattended.

> **IMPORTANT Password Confidentiality**
>
> Do not share your password with any colleague, including supervisors. Passwords must be changed at least every ninety days, and immediately if a compromise is suspected. Report any suspected unauthorised access to the IT Administrator within one business day.

B.1 Dashboard

The Dashboard is the default landing page after login. It provides a real-time snapshot of operations for the current business day. The data refreshes automatically every sixty seconds, so the figures shown reflect the latest information available without manual reloading.

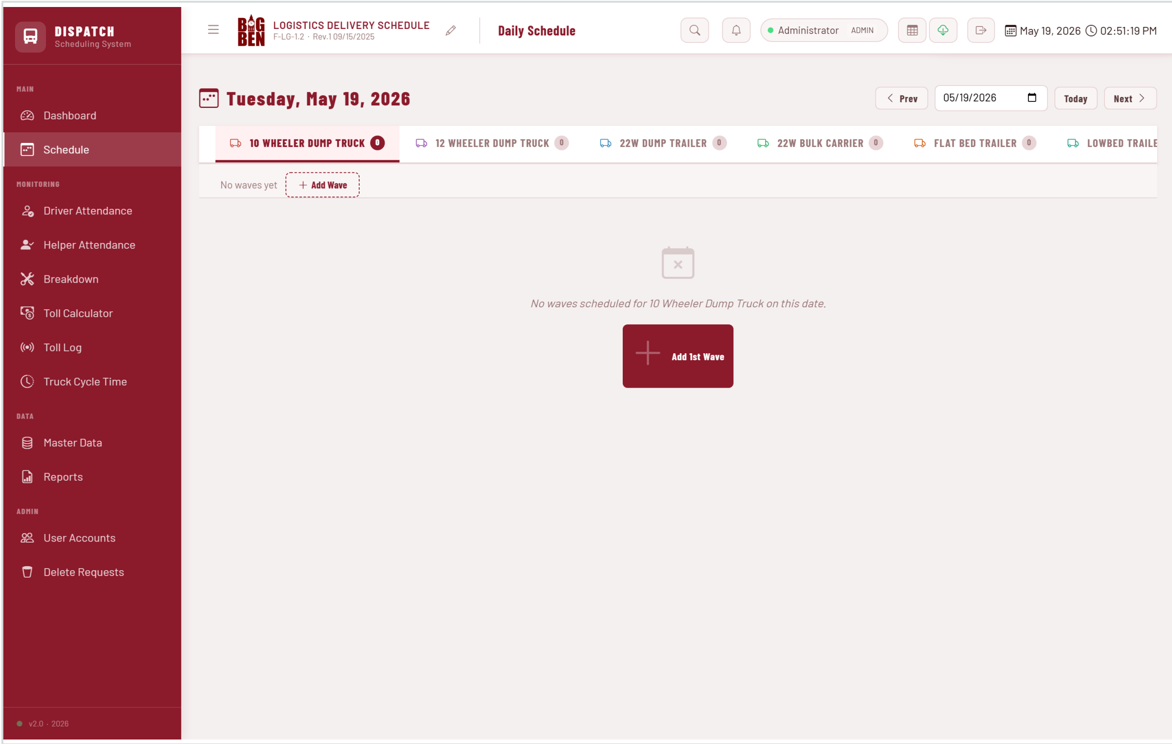

Figure: Dashboard overview — KPI cards, fleet utilisation, breakdown chart, truck cycle widget

KPI Cards

Four headline indicators sit at the top of the page. Each card is clickable and opens a drill-down dialog showing the underlying trips.

KPI	Definition	Drill-down
Trips Today	Count of TripRecord rows with a Wave date equal to today, excluding cancelled trips.	List of today's trips with status, plate, driver, client.
Toll Fee	Sum of toll_fee across today's non-cancelled trips. **Manual entry only** — dispatcher records each trip's toll from the physical receipt or RFID statement.	Same trip list with toll figures highlighted.

KPI	Definition	Drill-down
GPS Toll	Sum of toll fees computed by the polling worker from GPS plaza transits today. **Independent of Toll Fee** — purely reference / audit. Finance can compare the two to spot exemptions, missed transits, or RFID glitches.	Click the card to open the Toll Log page.
Delivered	Count of trips with status = Delivered.	Filtered list of completed trips only.
Fleet Util. %	Earned points divided by daily target points across all plates with at least one assignment.	Per-plate breakdown of points earned.

ISO 9001 REFERENCE Toll Fee vs GPS Toll

Two distinct KPIs by design. **Toll Fee** is what dispatchers manually record from receipts (this is what Finance bills). **GPS Toll** is what our polling worker auto-detected from plaza transits — it is a reference figure, not a billing input. Daily reconciliation (SOP-004) compares the two to catch discrepancies that warrant investigation (e.g., a transit the GPS detected but no manual entry exists, or vice-versa).

Fleet Utilisation Chart

The horizontal bar chart shows each truck type's percentage of daily target points achieved. Each truck type has a per-leg point value (typically 1.0, or 0.5 for ten-wheelers) and a daily target (typically 1.5, or 4.0 for ten-wheelers). Bars are colour-coded green (≥100%), amber (70–99%), and red (below 70%).

Driver-Truck Ratio

Compares the number of available drivers against the number of assigned plates per truck category. A ratio below 1.0 indicates plates outnumber qualified drivers and surfaces staffing gaps that require attention.

Breakdown Hours Chart

Stacked bar chart of total breakdown hours per plate over a configurable look-back window (default seven days). Used to identify chronically unreliable plates that may need preventative maintenance or replacement.

Truck Cycle Time Widget

Four mini-statistics: Open Cycles (trucks currently away from BIG BEN SCM), Closed Today, Average Cycle Hours over the past seven days, and Long Cycles (those exceeding twenty-four hours). Click the **Open Truck Cycle Time** button to jump to the dedicated cycle-time analytics page.

IMPORTANT Data Quality

Dashboard figures are only as accurate as the underlying trip data. Dispatchers must update trip statuses promptly. Trips left at "Pending" or "In Transit" after delivery cause the Delivered count to under-report. See SOP-002 (End-of-Day Reconciliation).

B.2 Schedule

The Schedule module is where the daily trip plan is built, updated, and closed out. It is the most heavily used module in the application and the primary tool of the dispatch team.

Layout

The Schedule page is organised by truck type. Each truck type has its own tab at the top of the page (e.g., DT, TH, MDT, L300). Selecting a tab shows all Waves scheduled for that truck type on the current date.

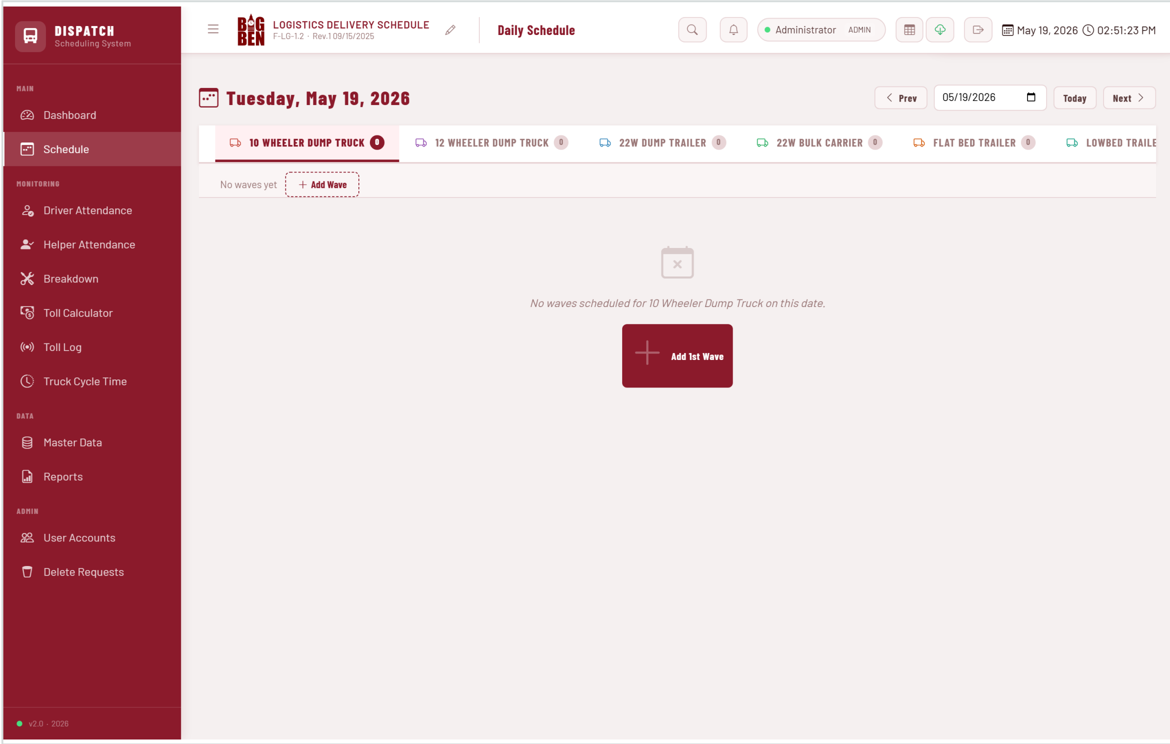

Figure: Schedule page — truck-type tabs, Wave panels with trip rows

Navigating Between Dates

Use the date picker at the top of the page. Selecting a new date reloads the schedule for that date without losing your current truck-type tab.

Creating a New Wave

A Wave is a grouping of trips on the same date and truck type. Most days have a single Wave per truck type, but multiple waves are supported (e.g., morning and afternoon dispatches).

1. Select the desired date and truck-type tab.

2. If no waves exist yet, click the **+ Add 1st Wave** button in the empty state.

3. If a wave already exists and you need a second, click the **+ New Wave** button at the top of the truck-type panel.

4. A wave label is automatically assigned (e.g., "Wave 1", "Wave 2").

Adding a Trip Row

1. Click **+ Add Row** at the bottom of the Wave panel.

2. A new editable row appears with empty fields.

3. Fill in the fields from left to right. Required fields: Driver, Plate, Product, Client. Optional but recommended: Helper, Dispatcher, RS No., PO No., DR No., Volume.

4. Set the Status — typically "Pending" at creation time.

5. Enter the Toll Fee if known in advance; otherwise leave at 0 and let the GPS auto-fill populate it.

6. Click anywhere outside the row to commit the changes. Saves occur automatically per field on blur.

Editing an Existing Trip

All fields are inline-editable. Click on the field, type or select the new value, then click outside the field. Changes save automatically. There is no "Save" button — this design eliminates the risk of losing edits.

Duplicating a Trip

Use the copy icon (■) at the start of each trip row to duplicate the row. The copy carries all field values (driver, helper, plate, product, client, dispatcher, RS/PO/DR numbers, volume, status, notes, toll fee) so dispatchers can rapidly create similar trips and tweak only the differences.

Duplicate to the same wave (default)

Single-click the copy icon. A new row appears at the bottom of the same wave, highlighted briefly in yellow. Edit the few fields that differ (typically RS / PO / DR numbers).

Duplicate to a different wave

When two or more waves exist for the truck type on the same date (e.g., morning and afternoon dispatches), the copy icon shows a small caret beside it. Click the caret to open a short menu listing every wave, with the current one marked "(here)". Pick the target wave to copy the row there. A toast confirms the destination (e.g., "Copied to 2nd Wave").

> **BEST PRACTICE** **Cross-wave copy speeds up multi-shift planning**
>
> Same client + product + plate + driver but split across AM and PM shifts is a common pattern. Use the cross-wave copy to clone the AM trip into the PM wave with one click; only the RS/PO/DR numbers need updating. See SOP-001 for the standard daily dispatch flow.

Updating Trip Status

The Status badge cycles through the four operational states when clicked: **Pending** → **Loading** → **In Transit** → **Delivered**. To set a non-linear status (e.g., Cancelled), use the dropdown control instead.

Status	Meaning	When to set
Pending	Trip planned but not yet started.	At trip creation, or when reverting an incorrect status.
Loading	Truck is at the origin loading cargo.	When the driver radios in at the loading point.
In Transit	Truck is en route to the customer.	When the truck departs the loading point.

Status	Meaning	When to set
Delivered	Cargo has been received and signed for.	On confirmation from the driver / customer.
Cancelled	Trip was assigned but did not run.	When a customer cancels, or weather / breakdown prevents execution.

Deleting a Trip

Click the red trash icon at the right end of the trip row. A confirmation dialog appears. Deletions are permanent and bypass the audit trail; prefer "Cancelled" status for trips that did not run for historical reasons.

> **IMPORTANT Deletion vs Cancellation**
>
> Use **Delete** only for rows created in error (typo, duplicate). For trips that were planned but did not run, set the status to **Cancelled** instead. This preserves the audit trail and surfaces operational issues in the cancellation rate KPI.

B.3 Master Data

The Master Data module manages the reference data used throughout the application: people (drivers, helpers, dispatchers), assets (plates, truck types), products, clients, and GPS mappings. Master data changes are infrequent but high-impact: an incorrect entry here propagates to every trip that references it.

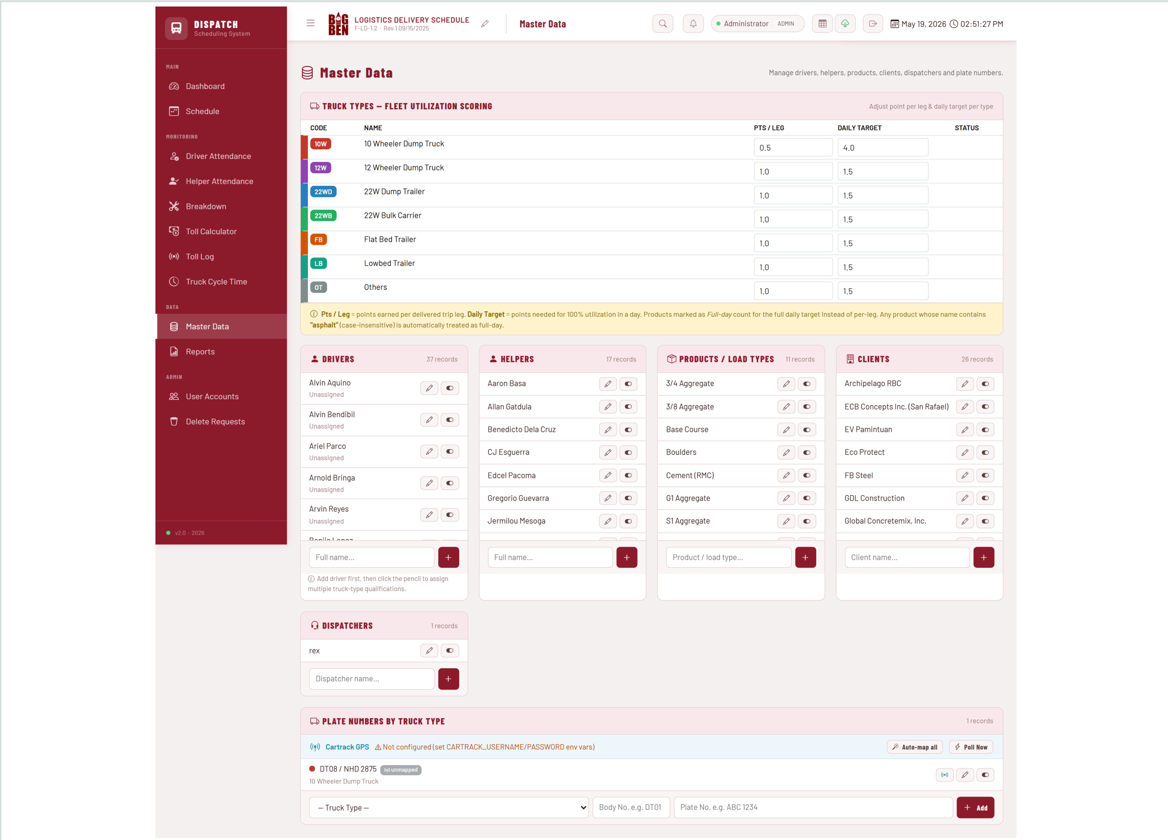

Figure: Master Data overview — cards for each entity type

Entity Cards

Each entity (drivers, helpers, plates, etc.) is displayed as a card with a header and a list of rows. Each card provides three actions: **Add**, **Edit**, and **Deactivate**.

Adding a Driver

1. Locate the **Drivers** card.

2. Click **+ Add Driver**.

3. Enter the driver's full name.

4. Select one or more truck types the driver is qualified to operate.

5. Confirm the **Active** toggle is on.

6. Click **Save**.

Adding a Plate

1. Locate the **Plate Numbers** card.

2. Click **+ Add Plate**.

3. Enter the registration plate number (e.g., NHF9508) in upper case.

4. Enter the body number (e.g., DT15) — the internal identifier.

5. Select the truck type from the dropdown.

6. Optionally map to a Cartrack vehicle ID for GPS tracking — see B.3.4.

7. Click **Save**.

Mapping Plates to Cartrack GPS Devices

For GPS-driven features (auto toll fill, cycle tracking, live status) to work, each plate must be linked to its corresponding Cartrack vehicle. The link is established by setting the plate's **cartrack_vehicle_id**.

1. Click the **Auto-map all** button in the Plates card toolbar. The system attempts to match each plate to a Cartrack vehicle by registration number.

2. Review the response — the system reports how many plates were matched, how many were ambiguous, and how many remained unmatched.

3. For unmatched plates, click the broadcast icon (■) beside the plate and pick the correct Cartrack vehicle from the dropdown.

4. Verify the cyan **CT #** badge appears beside the plate after mapping is complete.

Deactivating a Master Data Record

Historical records that are no longer in use should be **deactivated**, not deleted. Deactivation removes the entity from dropdowns on new trip rows but preserves it for historical reporting. To deactivate, toggle the **Active** switch on the entity row to off.

B.4 Breakdown

The Breakdown module records every event that removes a plate from operational service: mechanical failures, accident damage, scheduled maintenance, and other downtime causes. It is the primary record of fleet reliability.

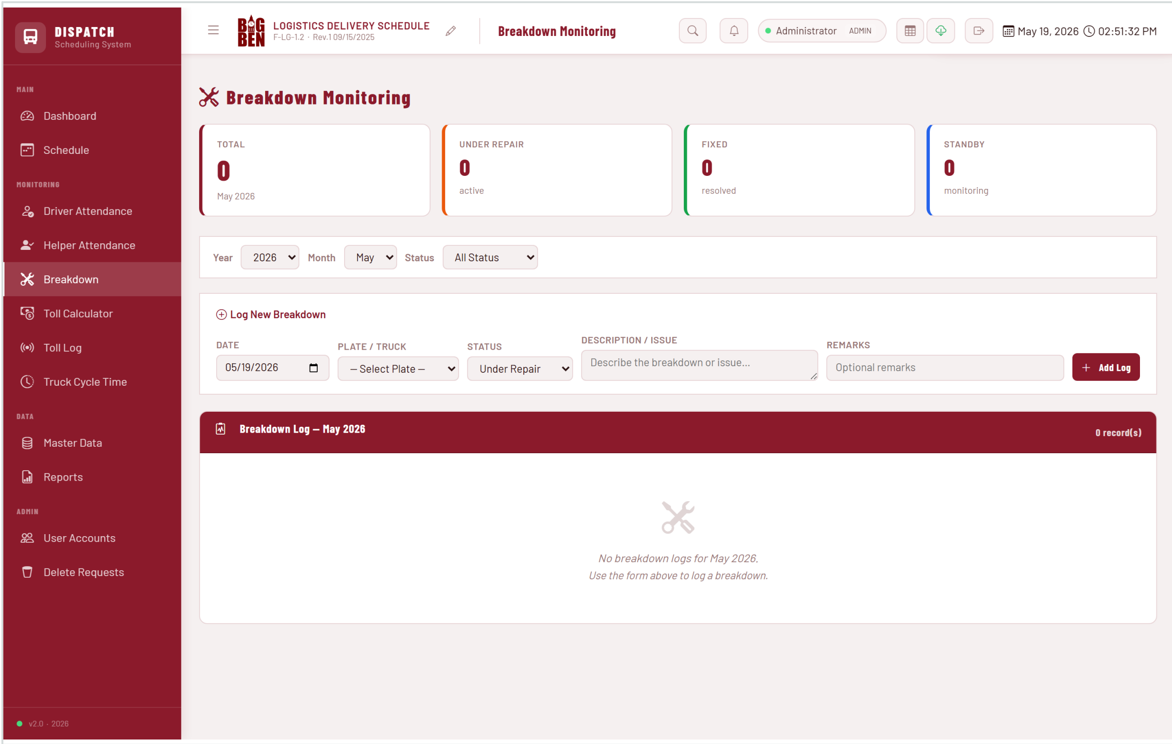

Figure: Breakdown module — incident table and log form

Logging a Breakdown

1. Navigate to **Breakdown** in the left sidebar.

2. Click **+ Log Breakdown**.

3. Select the affected plate.

4. Enter the start date and time of the incident.

5. Choose the status: **Under Repair**, **Fixed**, or **Standby**.

6. Enter a short description of the cause.

7. Optionally attach the resolution and the end date and time once known.

8. Click **Save**.

Updating Repair Status

Re-open the breakdown record. Change the status, enter the end date and time, and add the resolution description. Total downtime is calculated automatically as the difference between start and end timestamps.

Breakdown Categories

Status	Meaning
Under Repair	Plate is currently with a mechanic or in the shop. Not available for dispatch.
Fixed	Repair is complete. Plate is back in service.
Standby	Plate is operational but intentionally held out of service (e.g., waiting for parts, scheduled inspection).

> **ISO 9001 REFERENCE   Same-Day Reporting**
>
> Breakdowns must be logged within twenty-four hours of occurrence. Delayed reporting distorts fleet-availability metrics and complicates payroll calculations for affected drivers. See SOP-003 (Breakdown Reporting).

B.5 Toll Calculator

The Toll Calculator computes the expected toll fee between any two plazas on any supported expressway. It uses the same fee matrix that powers the GPS auto-fill feature, so manual and automatic figures are always consistent.

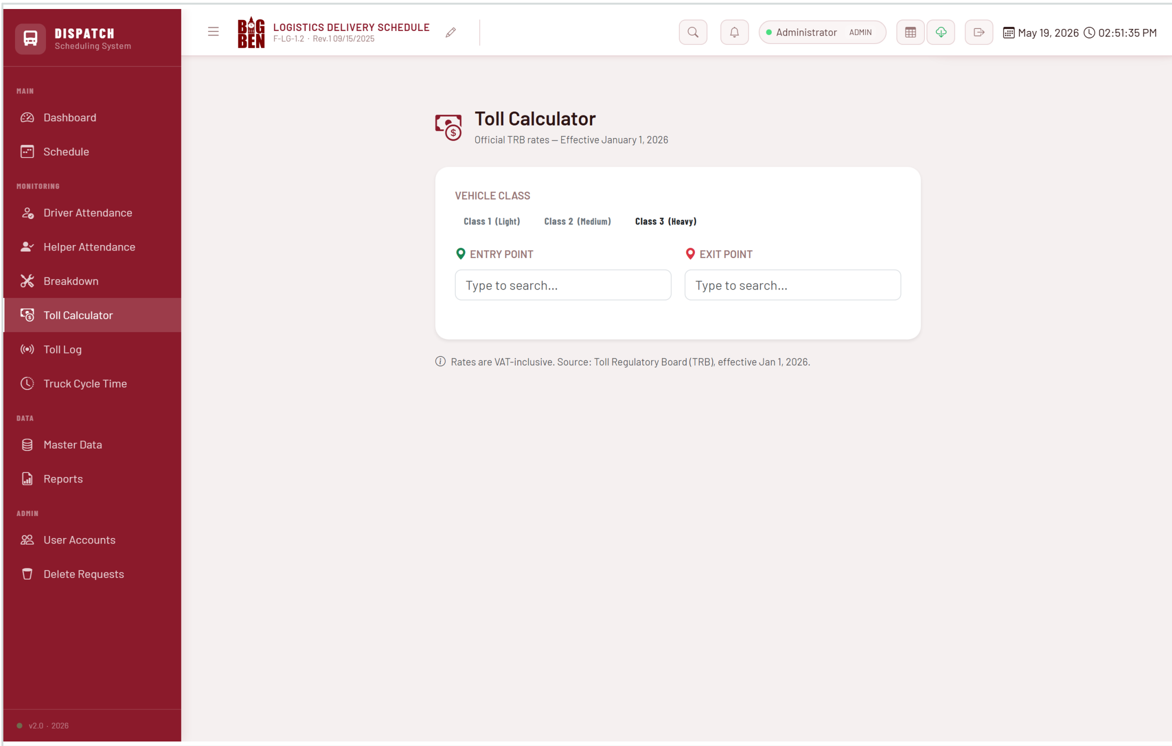

Figure: Toll Calculator — expressway selector, plaza dropdowns, computed fee display

Computing a Single-Expressway Fee

1. Choose the expressway from the dropdown (e.g., NLEX / SCTEX).

2. Choose the vehicle class (Class 1, 2, or 3).

3. Choose the entry plaza.

4. Choose the exit plaza.

5. The fee appears below the form, in Philippine pesos.

Multi-Expressway Routing

For trips that span multiple expressways (e.g., NLEX → SCTEX → CALAX), the calculator performs a graph search to find a valid route and sums the toll fees along the way. Intermediate plazas at expressway junctions are inferred automatically.

Supported Expressways

Expressway	# Plazas
NLEX / SCTEX	31
Skyway / SLEX / MCX	22
TPLEX	11
NAIAX	9
STAR Tollway	8
CALAX	7
Skyway Stage 3	7
CAVITEX	5
Harbor Link	5
NLEX Connector	3
Total	**108**

B.6 Toll Log

The Toll Log is a chronological list of every plaza-detection event captured by the GPS polling worker. Each event records the plate, expressway, plaza, event type (Enter or Exit), and timestamp. When the system pairs an Enter event with a subsequent Exit event at a different plaza, the matching trip is identified and its toll_fee is auto-filled.

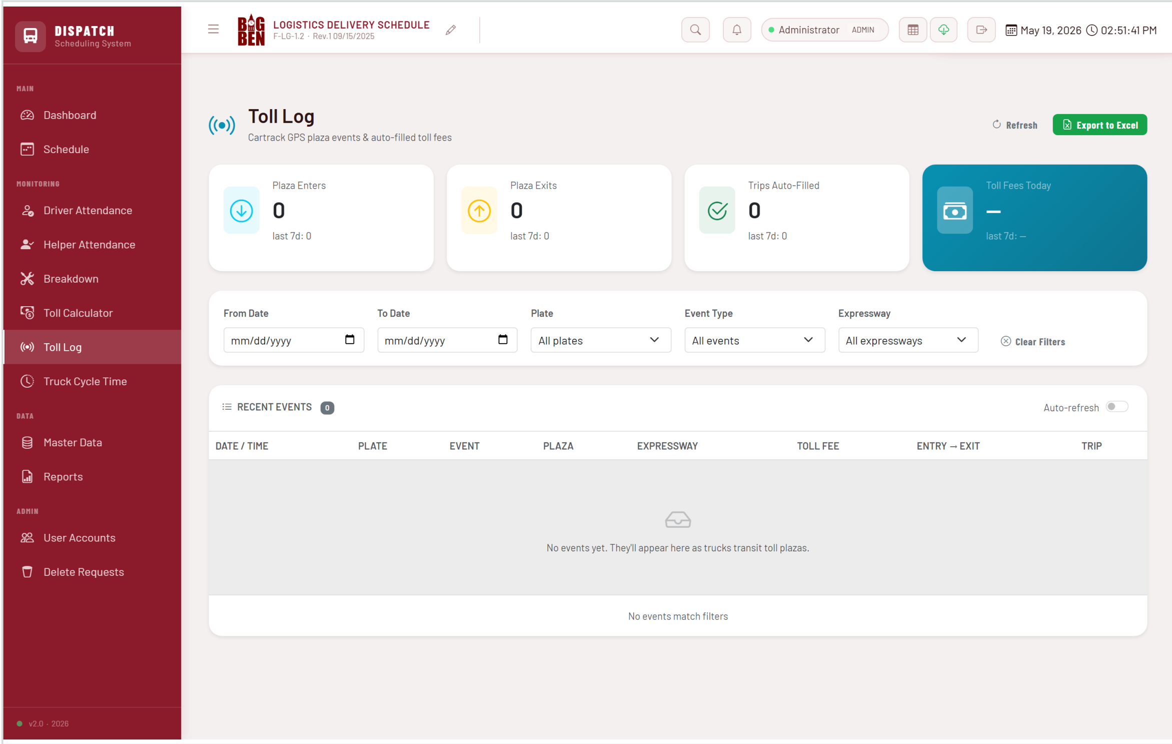

Figure: Toll Log page — KPI cards, filter panel, events table

KPI Cards

• **Plaza Enters** — count of ENTER events in the selected date range.

• **Plaza Exits** — count of EXIT events in the selected date range.

• **Trips Auto-Filled** — count of TripRecords whose toll_fee was populated by the GPS worker.

• **Toll Fees Today** — sum of auto-filled toll fees for the current day.

Filtering Events

Six filters are available: From Date, To Date, Plate, Event Type, Expressway, and a Clear button. Filters apply immediately upon change. Use them to investigate a specific transit or to audit a reported missing toll fee.

Exporting to Excel

Click **Export to Excel** to download all currently filtered events as an .xlsx file. The export preserves the filter state, so to export the full log, click **Clear Filters** first.

Reconciling Auto-Filled Tolls

Finance reconciles auto-filled tolls against physical receipts on a weekly basis. Any discrepancy greater than twenty pesos is investigated. Common causes of discrepancies are: a plate not mapped to Cartrack (no GPS evidence collected); a transit completed too quickly to be captured at the current polling cadence; or the plaza's GPS coordinates falling outside the actual booth location.

B.7 Truck Cycle Time

The Truck Cycle Time module is the fleet manager's live situational-awareness view. It shows, for every active plate, where the truck is right now, what it's doing, and the full audit trail of every stop and movement during its current and past round trips.

A **cycle** is one full round trip: the truck leaves home (BIG BEN SCM or any geofence with category=home), visits sites and toll plazas along the way, and returns to a home geofence. Cycles open the moment a truck exits a home geofence and close when it re-enters one. Multi-day cycles are supported.

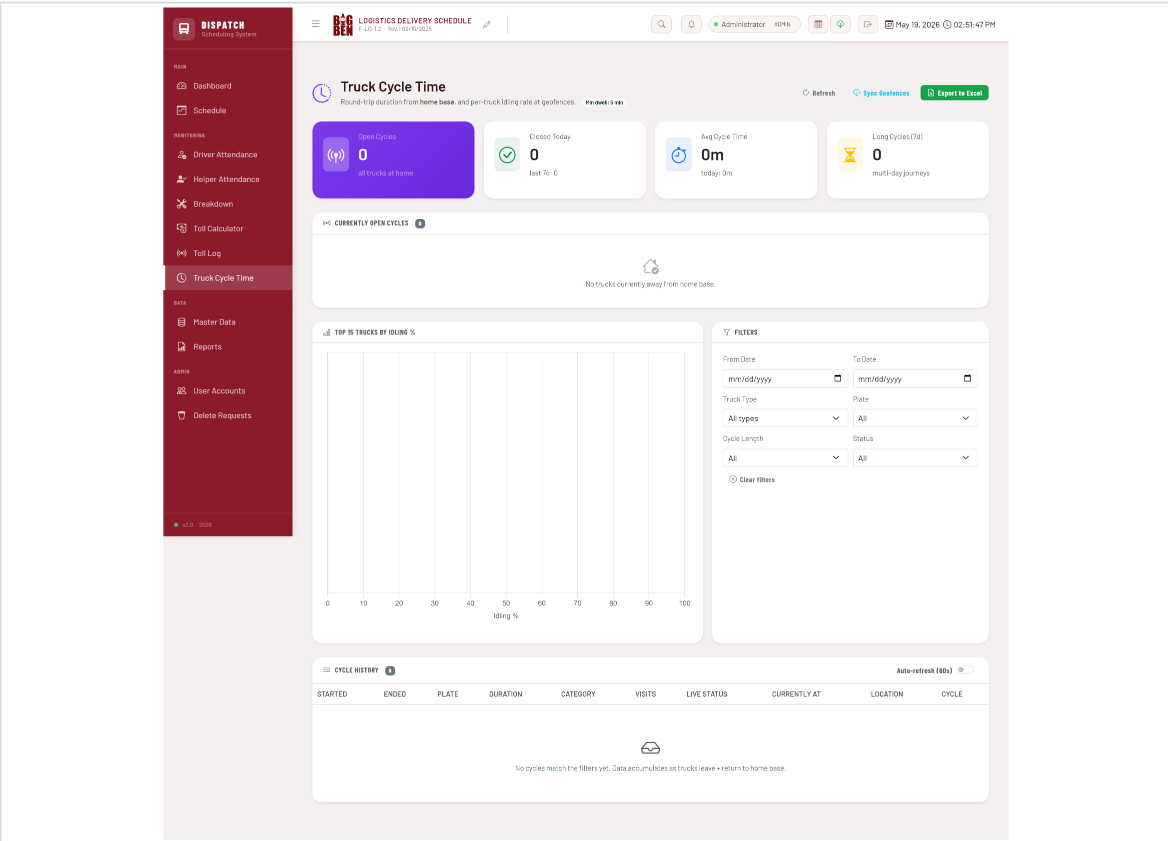

Figure: Truck Cycle Time — KPI cards, idling chart, plate-status table with expandable rows

KPI Cards

- **Open Cycles** — trucks currently away from home base.
- **Closed Today** — round trips that completed today.
- **Avg Cycle Time** — average round-trip duration over the past seven days.
- **Long Cycles** — round trips exceeding twenty-four hours over the past seven days.

Idling Rate Chart

A horizontal bar chart ranking the top fifteen trucks by overall idling percentage. Bars are colour-coded: green (under 15 percent), amber (15–30 percent), orange (30–50 percent), red (above 50 percent). Excessive idling indicates fuel waste or operational delays.

Filters

Five filters are available: From Date, To Date, **Truck Type**, Plate (cascades from truck type), Cycle Length (Short, Standard, Long, Ongoing), and Status (Open, Closed, All). Filters apply both to the Plate Status table and to the cycle picker inside an expanded row.

Plate Status Table

The page presents one row per active plate, not one row per cycle. Sorting is automatic: trucks with an open cycle appear at the top (longest elapsed first), and parked plates at the bottom (alphabetical, greyed out). At a glance the fleet manager can see exactly which trucks are on the road, how long they have been out, and where they last departed from.

Column	Description
Plate	Body number / registration (e.g., DT15 / NHF9508).
Status	Live status badge — DRIVING, IDLING, STOPPED, or OFF — colour-coded for quick scanning.
Currently At	Geofence name if the truck is inside one, "In transit" if on the road, or a real-time amber "Stopped — Xm" badge with the address if the truck is in an ongoing ad-hoc stop.
Last Departure	Time and location of the truck's most recent departure from a tracked location.
Out For	How long the open cycle has been running. Highlights amber when > 24 hours.
Stops	Count of real visits in the current open cycle (drive-bys excluded).
Location	Reverse-geocoded address from Cartrack — useful for trucks in transit between geofences.

Click a row to expand

Selecting any plate row reveals a panel below it with a chronological view of that plate's cycle history. Two controls appear:

• **Cycle picker chips** — one chip per cycle (open + closed) within the page's date filter range. The currently-open cycle is highlighted; closed cycles are grouped by date.

• **Audit trail timeline** — once a chip is picked, the panel renders the complete sequence of events for that cycle, in chronological order.

Audit trail event types

Icon	Kind	Source	Meaning
■	DEPARTED	SiteVisit	Truck left a known geofence (cycle home base, customer, quarry, etc.).
■	ARRIVED	SiteVisit	Truck entered a known geofence.
■■	STOPPED	SiteVisit	Ad-hoc stop — truck stopped for ≥ N minutes outside any geofence. Address captured from Cartrack reverse-geocode.
■■	PLAZA IN/OUT	CartrackEvent	Truck entered or exited a toll plaza geofence.
■	TOLL FILLED	CartrackEvent	GPS-detected toll fee computed (reference / Dashboard KPI — NOT written to the trip's Toll Fee field). See SOP-004.

Settings (Configurable Thresholds)

A **Settings** button in the page toolbar opens a modal where dispatch operators can adjust the two operational thresholds without code changes:

Setting	Default	Effect
Minimum visit dwell	5 min	Geofence visits shorter than this are flagged is_drive_by=True and hidden from analytics. Toll plazas are exempt — see SOP-004.
Stop detection threshold	10 min	Truck stopping outside any geofence for at least this duration is logged as an ad-hoc stop SiteVisit (with address). Below the threshold, the stop is ignored.

Both values are stored in the AppSetting table. The polling worker re-reads them on every iteration, so admin changes apply within the next polling cycle. Non-admin users see the settings as read-only.

Clear Logs (Admin Only)

Within the Settings modal, an admin-only Danger Zone exposes a Clear Logs action. This deletes tracking data older than a chosen cutoff date — useful when wiping trial-period noise before going fully live, or for periodic retention pruning.

What gets cleared

- **SiteVisit** rows (geofence visits + ad-hoc stops)
- **TruckCycle** rows (round-trip records)
- **CartrackEvent** rows (plaza ENTER / EXIT / trip_closed events)

What is preserved

- All master data (plates, drivers, helpers, products, clients, dispatchers, truck types)
- All TripRecord rows (including manual toll_fee entries)
- All BreakdownLog rows
- All CartrackGeofence rows (the geofence cache)

• All user accounts and AppSetting values

> **IMPORTANT Three independent safety gates**
>
> The Clear Logs action requires: (1) admin role at the API layer (non-admins get HTTP 403), (2) a cutoff date — full wipes are not allowed; only records older than the cutoff are deleted, and (3) the exact text **CLEAR** typed into a confirmation field. The submit button stays disabled until all three conditions are met, plus one final browser confirm() before the request fires.

Manual Geofences (Coord-Based Workaround)

Most geofences are managed in Cartrack and synced into the application via the **Sync Geofences** button. If a real Cartrack geofence ever becomes invisible to the API (e.g., a permission filter or pagination quirk), administrators can add an entry to `manual_geofences.json` at the project root. Each entry has a name, category, lat/lng, and radius in metres. The polling worker computes haversine distance from each truck's GPS to every manual geofence on every poll, and treats matches as if they came from Cartrack itself — same cycle and site-visit behaviour.

Categories supported: home, customer, quarry, fuel, toll, operations, other. Set category to "home" to make the geofence behave as a second home base for cycle open/close detection.

Cycle Categories

Category	Duration	Typical Pattern
Short	Under 12 hours	Single round trip within a single business day.
Standard	12 – 24 hours	Long single-day route, or a quick overnight.
Long	Over 24 hours	Multi-day journey (e.g., Cebu container runs).
Ongoing	Not yet closed	Cycle still open at the time the report is generated.

Syncing Geofences

Click **Sync Geofences** in the page toolbar to pull the latest geofence list from Cartrack. New geofences created in Cartrack are not visible to the system until this sync runs. The sync is safe to run on demand.

B.8 Reports

The Reports module produces summarised views over configurable date ranges. Reports are intended for management review and finance reconciliation rather than live operations.

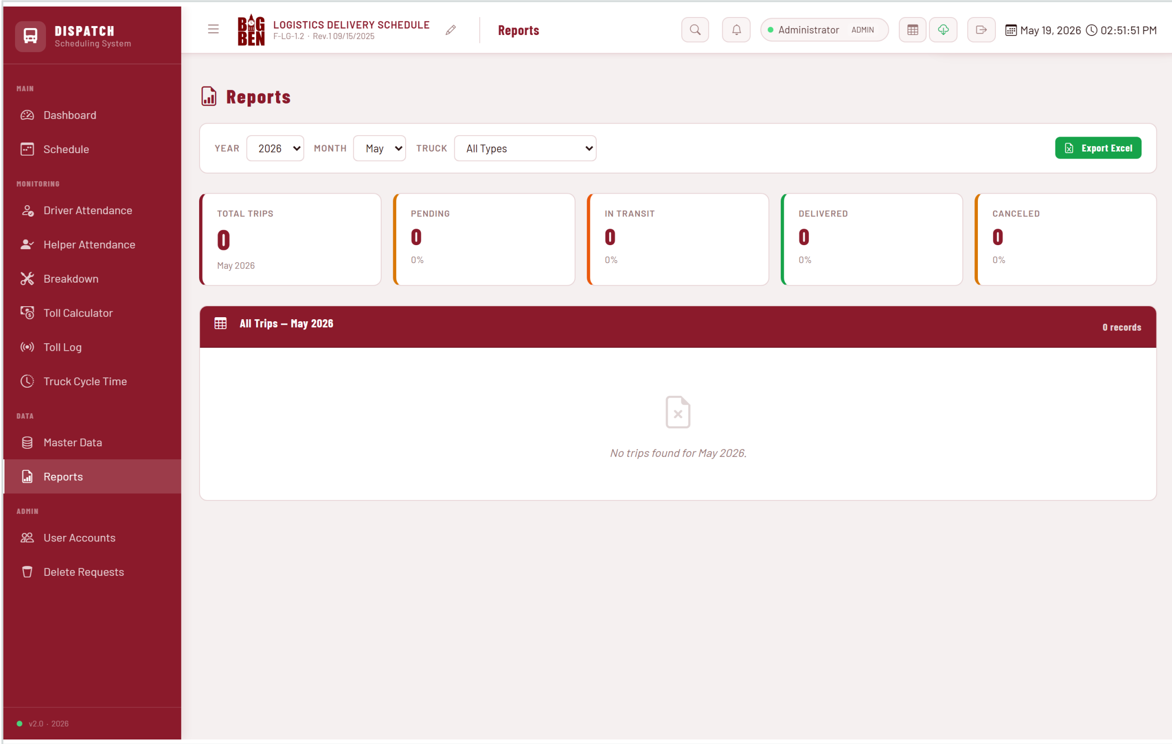

Figure: Reports page — date range, aggregate KPIs, trip table

Configurable Range

The date selector at the top of the page allows any continuous range. Common selections include: today, yesterday, this week, last week, this month, and last month.

Exports

Click **Export to Excel** to download the full trip set as an .xlsx file. Alternatively, click **Sync to Google Sheets** to push the data into the configured shared sheet for finance.

B.9 Administration and Settings

The Admin module is restricted by convention to the IT Administrator and the Management Representative. It provides access to user accounts, environment variables, integration configuration, and database maintenance utilities.

User Account Management

Create, deactivate, and reset passwords for application user accounts. Each user has a unique username and a hashed password; the system does not store plaintext passwords.

Environment Variables

Cartrack credentials, Google Sheets webhook URL, and other secrets are stored as environment variables on the PythonAnywhere host. They are not displayed in the application user interface. The IT Administrator manages them via the PythonAnywhere Web tab.

Database Backup

A scripted daily backup of the SQLite database is recommended. See SOP-005 (Data Backup and User Access Management) for the full procedure.

Section C — Standard Operating Procedures

This section documents the six core Standard Operating Procedures (SOPs) that govern the daily use of the Dispatch Scheduler System. Each SOP follows the ISO 9001:2015 format and is identified by a unique code (SOP-XXX) referenced throughout this manual.

ISO 9001 REFERENCE Mandatory Reading

All personnel who interact with the Dispatch Scheduler System are required to read and acknowledge the SOPs applicable to their role before operating the system unsupervised. Acknowledgement is recorded by the Management Representative in the Training Records (FM-001).

SOP-001

Daily Dispatch Planning and Execution

Field	Description
Purpose	To ensure that every operational day begins with a complete, approved, and executable trip plan, and that the plan is updated in the system promptly as conditions change.
Scope	Applies to all weekday and weekend dispatch operations conducted using the Dispatch Scheduler. Covers the period from the previous day's 6:00 PM planning meeting through the current day's end-of-day reconciliation.
Responsibilities	**Dispatcher** — builds the plan, assigns trips, updates statuses throughout the day. **Dispatch Supervisor** — reviews and approves the plan before 7:00 AM; intervenes on exceptions. **Operations Manager** — final accountability for plan completeness and execution.
Procedure	1. Receive confirmed customer purchase orders and trip requests from the previous day's planning meeting. 2. Open the Dispatch Scheduler and navigate to the **Schedule** module for the current operational date. 3. For each truck type, create the required Waves and add one trip row per planned delivery or pickup. Required fields: Driver, Helper, Plate, Product, Client, Dispatcher. 4. Cross-check that each assigned Plate is not currently recorded as Under Repair in the Breakdown module. If it is, reassign to an available Plate. 5. Cross-check that each assigned Driver is qualified for the Plate's Truck Type. Master Data enforces this implicitly via the Driver-Type relationship; resolve any mismatch before proceeding. 6. Enter the expected toll fee per trip using the Toll Calculator. Trips that auto-fill from GPS will overwrite this figure later. 7. Submit the day's plan for Dispatch Supervisor review. 8. On supervisor approval, set the trip statuses to **Pending** (default) — no further action required until dispatch. 9. As trucks depart, update trip status to **Loading** upon arrival at loading point, then **In Transit** on departure. 10. On delivery confirmation, update status to **Delivered** and verify the toll_fee has populated. If still zero, investigate (see SOP-004).
Records Generated	TripRecord rows in the database; Wave records; status change timestamps (audit log); auto-fill toll evidence (CartrackEvent + SiteVisit rows).
Frequency	Daily — every operational day, including weekends.
References	SOP-002 End-of-Day Reconciliation SOP-004 Toll Documentation ISO 9001:2015 Clause 8.5 (Production and service provision)

SOP-002

End-of-Day Reconciliation

Field	Description
Purpose	To ensure that every trip planned for the operational day has been closed out with an accurate final status, that all toll fees have been verified, and that any unresolved exceptions are documented and escalated before the daily records are considered final.
Scope	Performed at the end of each operational day, after all trucks have either returned to BIG BEN SCM or been confirmed as multi-day cycles in progress.
Responsibilities	**Dispatcher (closing shift)** — performs the reconciliation. **Dispatch Supervisor** — reviews and signs off. **Finance Officer** — receives the closed-out totals for next-day reconciliation.
Procedure	1. Navigate to the **Schedule** module for the current date. 2. Iterate through each truck-type tab in turn. 3. For each trip row, confirm the status is one of: **Delivered**, **Cancelled**. Any trip still in **Pending**, **Loading**, or **In Transit** is an exception. 4. For each exception, contact the assigned driver or dispatcher to determine the actual outcome. Update the status accordingly. 5. Open the **Toll Log** page and filter to the current date. 6. For each Delivered trip, verify a corresponding toll auto-fill is present, or that the manual toll entry matches the physical receipt. 7. For trips with toll_fee = 0 that should have a toll, manually enter the figure from the receipt. Add a note in the trip's Notes field indicating "Manual entry — no GPS evidence". 8. Open the **Dashboard** and confirm the Trips Today, Delivered, and Total Toll KPIs reflect the expected figures. 9. Click **Sync to Google Sheets** on the Reports page to push the day's data for Finance. 10. Notify the Dispatch Supervisor that reconciliation is complete.
Records Generated	Final TripRecord statuses; manually entered toll fees with Notes; Google Sheets sync timestamp; Supervisor sign-off (verbal or written log).
Frequency	Daily — within two hours of the last truck returning to home base.
References	SOP-001 Daily Dispatch SOP-004 Toll Documentation ISO 9001:2015 Clause 8.6 (Release of products and services)

SOP-003

Breakdown Reporting and Plate Availability Management

Field	Description
Purpose	To ensure that every mechanical, electrical, or accident-related event that takes a plate out of service is recorded promptly, the affected plate is removed from dispatch availability, and the operational impact is communicated to all relevant parties.
Scope	Applies to any event that prevents a plate from completing its planned trip or being assigned to subsequent trips. Includes scheduled maintenance, unplanned breakdowns, accidents, and standby holds.
Responsibilities	**Driver** — reports the event immediately to the Dispatcher. **Dispatcher** — opens a Breakdown record in the system within thirty minutes of notification. **Fleet Manager** — assigns the plate to a mechanic, tracks repair progress, updates the resolution.
Procedure	1. Upon driver notification of a breakdown, record the time, location, and reported symptoms. 2. Open the **Breakdown** module in the Dispatch Scheduler. 3. Click **+ Log Breakdown** and complete the form: Plate, Start Date / Time, Status = **Under Repair**, Description (driver's reported symptoms). 4. Save the breakdown record. The plate is now flagged as unavailable in dispatch dropdowns. 5. Reassign the affected trip's Plate field to an available plate of the same truck type. If none is available, escalate to the Dispatch Supervisor. 6. Notify the Fleet Manager via Viber or radio for recovery / repair dispatch. 7. On receipt of repair confirmation from the mechanic, re-open the breakdown record and set Status = **Fixed**, enter the End Date / Time, and add the resolution description. 8. Confirm the plate is back in the dispatch pool for the next operational day.
Records Generated	BreakdownLog rows in the database; total downtime calculated automatically; Fleet Manager's mechanic communication log.
Frequency	Event-driven — within thirty minutes of any service-affecting incident.
References	SOP-001 Daily Dispatch ISO 9001:2015 Clause 7.1.3 (Infrastructure) ISO 9001:2015 Clause 8.7 (Control of nonconforming outputs)

SOP-004

Toll Documentation and Reconciliation

Field	Description
Purpose	To ensure that every toll fee incurred by the fleet is recorded accurately and that the manual entries on trip records (the billing source of truth) reconcile with the physical evidence (toll receipts, RFID statements) and with the system's GPS-detected toll figures (reference).
Scope	All trips that traverse a Philippine expressway with a toll plaza. Excludes purely intra-city trips that incur no toll.
Responsibilities	**Dispatcher** — manually records the toll_fee on each trip from the physical receipt at end of trip. **Finance Officer** — reconciles weekly against the GPS Toll Dashboard KPI, RFID statements, and physical receipts. **Fleet Manager** — investigates any reconciliation discrepancy greater than twenty pesos.
Procedure	1. At end of trip, the dispatcher collects the physical toll receipt from the driver and enters the toll fee directly into the trip's Toll Fee field on the Schedule page. This is the **sole source of truth** for Finance billing. The polling worker does NOT modify this field. 2. Throughout the day, the GPS polling worker detects plaza transits and computes the toll fee from the rate matrix. These figures are logged as CartrackEvent rows and aggregated into the Dashboard's **GPS Toll** KPI card. They are **reference / audit only** — they do not overwrite the dispatcher's manual entry. 3. At end-of-day reconciliation (SOP-002), the dispatch supervisor cross-checks the day's Toll Fee total (manual) against the GPS Toll figure on the Dashboard. A small variance is expected; large gaps trigger investigation via the Toll Log page. 4. On a weekly cadence (every Monday), Finance opens the **Toll Log** module, filters to the prior week, and exports to Excel. 5. Finance produces a three-way reconciliation: TripRecord.toll_fee totals (manual), CartrackEvent totals (GPS), and RFID/receipt totals (physical). Discrepancies above twenty pesos per trip are flagged. 6. Resolution of each discrepancy is recorded in the trip's Notes field and signed off by the Fleet Manager. Common causes: a toll exemption, a missed plaza (hidden geofence — see Manual Geofences in B.7), or a receipt typo.
Records Generated	TripRecord.toll_fee values (manual, the billing record); CartrackEvent rows with toll_fee + plaza pair (GPS reference); weekly reconciliation Excel exports; physical receipts (paper file).
Frequency	Per trip (manual entry at end of trip); end-of-day (per SOP-002); weekly (Finance reconciliation, every Monday).
References	SOP-001 Daily Dispatch SOP-002 End-of-Day Reconciliation Manual v1.1 — Toll auto-fill decoupling change ISO 9001:2015 Clause 7.1.5 (Monitoring and measuring resources)

SOP-005

Data Backup and User Access Management

Field	Description
Purpose	To protect the integrity and availability of the operational records held in the Dispatch Scheduler database, and to ensure that user access is granted, maintained, and revoked in line with the principle of least privilege.
Scope	Applies to the SQLite database file, the application codebase, all environment variables, and every user account on the system.
Responsibilities	**IT Administrator** — performs backups, manages user accounts, holds the recovery credentials. **Operations Manager** — initiates account requests and revocations. **Management Representative** — reviews access list quarterly.
Procedure	1. **Backups.** A daily scheduled task on PythonAnywhere copies the SQLite database to a date-stamped file in a backups directory. The seven most recent daily backups are retained. 2. **Off-site copy.** Each Friday, the IT Administrator downloads the latest daily backup to a designated encrypted external storage location. 3. **Restore drill.** Once per quarter, the IT Administrator restores the most recent backup to a staging environment and verifies the database opens cleanly with no integrity errors. 4. **Account request.** The Operations Manager sends a written request to the IT Administrator naming the new user and their role. 5. **Account creation.** The IT Administrator creates the account with a randomly generated initial password, communicated to the new user out-of-band. The user is required to change the password on first login. 6. **Periodic review.** The Management Representative reviews the active user list every quarter and confirms each account is still required. 7. **Account revocation.** On notification of staff separation, the IT Administrator deactivates the account within one business day. Account data is retained for audit purposes; the account is not deleted.
Records Generated	Daily backup files (date-stamped); restore-drill log; account creation and revocation log; quarterly access review minutes.
Frequency	Daily (backup); weekly (off-site copy); quarterly (restore drill, access review); event-driven (account creation, revocation).
References	ISO 9001:2015 Clause 7.5.3 (Control of documented information) ISO 9001:2015 Clause 7.1.3 (Infrastructure)

SOP-006

Cycle Time and Idling Rate Monitoring

Field	Description
Purpose	To use the GPS-driven Truck Cycle Time analytics to identify operational inefficiencies, address chronic idling, and surface long cycles that may indicate underlying problems with routing, customer wait times, or vehicle reliability.
Scope	Performed weekly on the consolidated cycle-time data for the prior calendar week. Covers all plates that completed at least one cycle in the period.
Responsibilities	**Fleet Manager** — runs the weekly review and produces the briefing note. **Operations Manager** — reviews the briefing and approves any operational adjustments. **Dispatch Supervisor** — implements approved adjustments in the following week's plan.
Procedure	1. Every Monday morning, open the **Truck Cycle Time** module. 2. Set the date filter to cover the prior calendar week (Monday 00:00 to Sunday 23:59). 3. Note the four KPI values: Open Cycles, Closed Last Week, Avg Cycle Time, Long Cycles. 4. Review the Idling Rate chart. Investigate any truck with an idling percentage above thirty percent — talk to the driver, review the visit log, and identify whether the cause is operational (long customer waits) or behavioural (unnecessary engine running). 5. Review the Cycle History table, sorted by Duration descending. For each cycle exceeding twenty-four hours, verify the multi-day journey was planned (e.g., Cebu route) and not the result of an unrecognised breakdown or detour. 6. Compile a one-page briefing note for the Operations Manager. Include: KPI summary, top three idling offenders, top three long cycles, and recommended actions. 7. Discuss findings at the Monday operations meeting. Document approved actions in the meeting minutes. 8. Implement approved actions in the following week's planning.
Records Generated	TruckCycle rows in the database; SiteVisit rows linked to each cycle; weekly briefing notes (filed by the Fleet Manager); operations meeting minutes.
Frequency	Weekly — every Monday morning.
References	ISO 9001:2015 Clause 9.1 (Monitoring, measurement, analysis and evaluation) ISO 9001:2015 Clause 10.3 (Continual improvement)

Section D — Quality Management

This section maps the operational use of the Dispatch Scheduler System to the requirements of ISO 9001:2015. It describes how the system supports quality control, change management, access control, auditability, and continual improvement.

D.1 Data Quality Controls

Operational decisions are only as good as the data on which they are based. The system enforces data quality through a combination of validation, defaults, and reconciliation procedures.

Control	Mechanism	Owner
Required field validation	The application rejects trip rows missing Driver, Plate, Product, or Client.	Dispatcher
Type-qualified driver assignment	Driver dropdowns on trip rows are filtered to drivers qualified for the truck type of the Wave.	System
Plate availability check	Plates currently flagged as Under Repair are hidden from the Plate dropdown on new trip rows.	System / Fleet Manager
Status workflow integrity	Trip status cycles through Pending → Loading → In Transit → Delivered. Backward transitions require explicit dropdown selection.	Dispatcher
GPS evidence for tolls	Auto-fills carry their source CartrackEvent and SiteVisit IDs for traceability.	System
Daily reconciliation	Every Delivered trip must have either an auto-filled toll or a manually entered toll with a Notes annotation by end-of-day.	Dispatcher
Weekly toll reconciliation	Finance compares system toll figures against physical receipts and RFID statements every Monday.	Finance
Quarterly access review	Management Representative confirms each active user account is still required.	Management Rep.

D.2 Change Management

Changes to the application — whether new modules, bug fixes, or master-data restructuring — follow a controlled release process to prevent operational disruption.

Change Categories

Category	Examples	Approval
Master Data	Adding a new client, new truck type, new driver.	Operations Manager
Configuration	Adjusting poll cadence, drive-by threshold, cycle hard-close window.	IT Administrator + Ops Manager
Code change	New module, bug fix, UI enhancement.	IT Administrator + Management Representative
Schema migration	Adding a database column, new entity type.	IT Administrator + Management Representative
Vendor integration	New Cartrack endpoint, Google Sheets webhook change.	IT Administrator + Ops Manager

Release Procedure

1. Proposed changes are documented in writing (email, ticket, or change request form).

2. Changes are developed and tested in a separate Git branch.

3. Approval is obtained from the appropriate role(s) per the category table above.

4. Code is merged to the main branch and tagged with a version number.

5. Deployment to production is performed during a maintenance window, typically 6:00 AM on a non-operational day.

6. The Revision History on page 2 of this manual is updated, and a new version of the manual is distributed.

7. Post-deployment, the IT Administrator monitors the system for one hour to confirm normal operation.

> **IMPORTANT Roll-back**
>
> Every code release is preceded by a database backup (SOP-005). In the event of a critical post-release defect, the IT Administrator may roll back to the prior code version and, if necessary, restore the pre-release database backup. Roll-backs are reported to the Management Representative within twenty-four hours.

D.3 Access Control

Access to the Dispatch Scheduler is granted on the principle of least privilege. There is no anonymous access — every request to every page is authenticated against the user database.

- Each user has a unique username. Shared accounts are prohibited.
- Passwords are hashed using a one-way cryptographic function. The plaintext password is never stored.
- A new user's initial password is randomly generated and communicated out-of-band; the user must change it on first login.
- Passwords expire every ninety days and must be changed.

• Sessions expire automatically after twelve hours of inactivity.

• Failed login attempts are logged. Five consecutive failures against a single account trigger a fifteen-minute lockout.

• The IT Administrator reviews the active user list quarterly, in coordination with HR.

D.4 Audit Trail

The system maintains an audit trail across the operational data sufficient to reconstruct what happened, when, and to which records, in support of internal and external audits.

Audit Element	Captured? (Y/N)	Retention
Trip record creation timestamp	Y	Indefinite
Trip record field changes (full history)	Partial (last update timestamp)	Indefinite
Trip status transitions	Y (timestamped per change)	Indefinite
Login successes / failures	Y (via web-server logs)	60 days
Breakdown record creation / closure	Y	Indefinite
Toll auto-fill events	Y (via CartrackEvent + SiteVisit)	60 days for events, indefinite for the resulting TripRecord.toll_fee
Cartrack polling worker runs	Y (always-on task log)	60 days
Database backups	Y (daily file snapshots)	7 daily + weekly off-site

D.5 Backup and Recovery

Operational continuity is protected by a layered backup strategy. See SOP-005 for the full procedure. Recovery objectives are stated below.

Metric	Target
Recovery Time Objective (RTO)	4 hours — from incident detection to operational restore.
Recovery Point Objective (RPO)	24 hours — at worst, one day of operational data is lost.
Backup Frequency	Daily (automated); Weekly (off-site copy).
Backup Retention	7 daily on-host + indefinite weekly off-site.
Restore Testing	Quarterly drill, documented in the restore-drill log.

Section E — Technical Documentation

This section provides technical reference material for the IT Administrator and external auditors. It complements the operational sections by documenting the system's internal structure: data model, API endpoints, integration points, and deployment topology.

E.1 Deployment Topology

The production deployment runs on PythonAnywhere under the paid Developer tier. The web application is served by PythonAnywhere's WSGI server. A separate always-on task runs the GPS polling worker. Both processes read the same environment variables and the same SQLite database file.

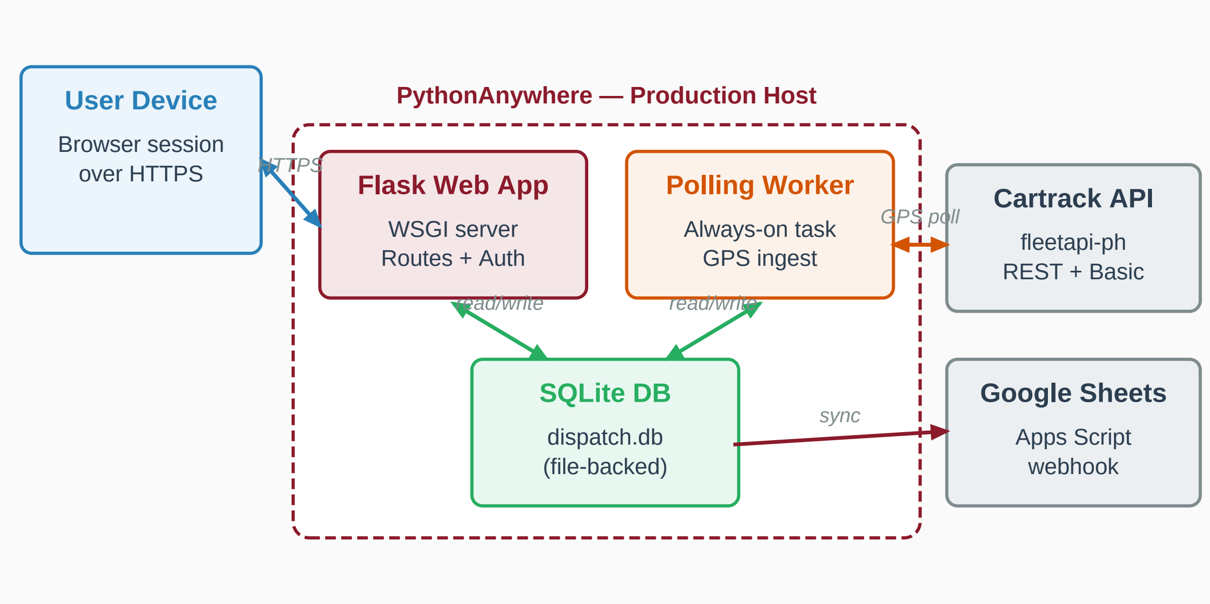

Figure E-1: Deployment topology. Both the Flask web application and the GPS polling worker run as separate processes on the PythonAnywhere host and share the SQLite database. The Cartrack API and Google Sheets webhook are external HTTPS dependencies.

Process List

Process	Purpose	Trigger
Web application (Flask)	Serves user-facing pages and JSON APIs.	PythonAnywhere WSGI on inbound HTTPS request.
Polling worker (cartrack_poll.py)	Fetches GPS positions, geofence events, processes site visits, opens/closes cycles, auto-fills tolls.	PythonAnywhere always-on task, restart on host reboot.
Database (SQLite)	Single-file relational store.	Always available (file-backed).

E.2 Database Schema (Key Entities)

The database uses SQLite via SQLAlchemy ORM. The principal entities and their relationships are summarised below. Full schema definitions live in `models_v2.py`.

Entity	Purpose	Key Relations
TruckTypeDef	Truck type catalogue (DT, TH, MDT, …).	Has many Plates, Waves.
Driver	Personnel who drives trucks.	Qualified for one or more TruckTypeDefs.
Helper	Personnel who assists drivers.	Independent — assignable to any trip.
Plate	Vehicle in the fleet.	Belongs to one TruckTypeDef. May map to one Cartrack vehicle.
Client	Customer / consignee.	Used in trip records.
Product	Material being transported.	Used in trip records.
Wave	Group of trips on one date for one truck type.	Has many TripRecords.
TripRecord	A single delivery / pickup assignment.	References Wave, Driver, Helper, Plate, Product, Client.
BreakdownLog	Plate-out-of-service event.	References one Plate.
CartrackTruckState	Latest GPS snapshot per plate.	One-to-one with Plate.
CartrackGeofence	Geofence imported from Cartrack.	Categorised: home, customer, quarry, toll, …
CartrackEvent	Raw GPS event (enter / exit a geofence or plaza).	References Plate, optionally CartrackGeofence.
SiteVisit	Aggregated visit to a geofence (enter + exit pair).	References Plate, CartrackGeofence, optionally TripRecord and TruckCycle.
TruckCycle	One round trip from home base and back.	References Plate. Has many SiteVisits.

E.3 Selected API Endpoints

The application exposes a number of JSON endpoints used internally by the front-end. They are documented here as reference for the IT Administrator. All endpoints require an authenticated session.

Endpoint	Method	Purpose
/api/dashboard/kpis	GET	Headline KPI figures (now includes GPS Toll aggregate).
/api/dashboard/fleet-utilization	GET	Per-truck-type utilisation.
/api/dashboard/driver-truck-ratio	GET	Driver vs assigned plate ratio.
/api/dashboard/breakdown-hours	GET	Per-plate downtime histogram.

Endpoint	Method	Purpose
/api/cycle-time/summary	GET	Cycle KPIs + open-cycle list.
/api/cycle-time/cycles	GET	Filtered cycle history (legacy, used by Excel export).
/api/cycle-time/plates	GET	Plate-centric live view used by the redesigned Plate Status table.
/api/cycle-time/plate/<id>/cycles	GET	List of cycles for one plate — feeds the cycle-picker chips when a row is expanded.
/api/cycle-time/cycle/<id>/timeline	GET	Full chronological audit trail of one cycle (visits + plaza events + in-progress stops).
/api/cycle-time/idling	GET	Per-plate idling statistics.
/api/cycle-time/filters	GET	Dropdown options for the page.
/api/cycle-time/export	GET	Excel export of cycles.
/api/cycle-time/sync-geofences	POST	Admin trigger to pull geofences from Cartrack (uses limit=1000).
/api/cycle-time/settings	GET	Read the current min-visit and stop-detection thresholds.
/api/cycle-time/settings	POST	Admin-only: update the thresholds (persists to AppSetting; polling worker picks up on next poll).
/api/cycle-time/clear-logs	POST	Admin-only: purge tracking rows (SiteVisit, TruckCycle, CartrackEvent) older than a required cutoff date. Requires confirm="CLEAR".
/api/toll-log/summary	GET	Toll Log KPI figures.
/api/toll-log/events	GET	Filtered list of plaza events.
/api/toll-log/export	GET	Excel export of events.
/api/cartrack/status	GET	Diagnostic status of the GPS integration.
/api/cartrack/poll-now	POST	Manually trigger a polling cycle.
/api/cartrack/auto-map	POST	Auto-link plates to Cartrack vehicles.
/api/trip/save	POST	Create / duplicate a trip. The Schedule cross-wave copy dropdown uses this with a target wave_id.
/api/sync-to-sheets	POST	Push operational data to Google Sheets.

E.4 External Integrations

Cartrack Fleet REST API

Endpoint host: `fleetapi-ph.cartrack.com`. Authentication uses HTTP Basic with credentials stored in environment variables (CARTRACK_USERNAME, CARTRACK_PASSWORD). The polling worker fetches per-vehicle status approximately once per minute and processes the geofence-id list and idling / ignition flags to update local state.

Google Sheets Webhook

When configured, the application can push trip and breakdown data to a Google Sheet via a Google Apps Script webhook. The URL is stored in the AppSetting table and is invoked from the /api/sync-to-sheets endpoint. Push is one-way (no read-back).

Section F — Appendices

Reference material that supports the operational and quality sections of this manual: troubleshooting, forms, contact information, and a quick-reference cheat sheet.

F.1 Troubleshooting Guide

Common operational issues and their resolutions.

Symptom	Probable Cause	Resolution
Cannot log in — "invalid credentials"	Password expired, mistyped, or account deactivated.	Try again carefully. If still failing, contact the IT Administrator for a password reset.
Trip toll_fee shows 0 after delivery	GPS evidence not captured — plate not mapped to Cartrack, plaza missed by the polling worker, or trip outside the date-match window.	Enter the toll manually from the physical receipt. Add a Notes annotation. Investigate per SOP-004.
Truck shows live status "NO DATA"	Plate not mapped to Cartrack, or the Cartrack device is offline.	Confirm Cartrack mapping in Master Data. Check the device in the Cartrack Fleet Web map.
Cycle stays "ongoing" after truck returns	Polling worker missed the home-geofence re-entry; or the cycle is genuinely multi-day.	Wait for the next polling cycle (one minute). If still unresolved after one hour, alert the IT Administrator.
Dashboard KPIs differ from expected totals	Trips left in Pending / Loading / In Transit are excluded from Delivered counts.	Close out remaining trip statuses per SOP-002.
Cartrack diagnostic shows "not configured"	Environment variables not set on the always-on task.	IT Administrator: confirm CARTRACK_USERNAME and CARTRACK_PASSWORD are set in both the Web tab and the always-on task environment.
Google Sheets sync returns an error	Webhook URL mis-configured or expired Apps Script trigger.	Verify the webhook URL in Admin / Settings. Re-publish the Google Apps Script if needed.

F.2 Forms and Records

The following forms support the SOPs. Each form is identified by a unique code (FM-XXX) and maintained by the Management Representative.

Code	Title	Used in SOP	Custodian
FM-001	Training Acknowledgement Record	All SOPs	Mgmt. Rep.
FM-002	Daily Dispatch Plan (printed copy)	SOP-001	Dispatch Supervisor

Code	Title	Used in SOP	Custodian
FM-003	End-of-Day Reconciliation Checklist	SOP-002	Dispatcher
FM-004	Breakdown Notification Slip	SOP-003	Fleet Manager
FM-005	Weekly Toll Reconciliation Worksheet	SOP-004	Finance Officer
FM-006	Backup / Restore Drill Log	SOP-005	IT Administrator
FM-007	User Access Request / Revocation	SOP-005	IT Administrator
FM-008	Weekly Cycle Time Briefing	SOP-006	Fleet Manager

F.3 Quick Reference — Common Tasks

Task	Module	Steps
Add a new trip	Schedule	+ Add Row → fill fields → blur to save.
Duplicate a trip (same wave)	Schedule	Click the copy icon (■) on the row.
Duplicate a trip to another wave	Schedule	Click the caret (▼) next to the copy icon → pick target wave.
Cancel a trip	Schedule	Click Status badge dropdown → Cancelled.
Log a breakdown	Breakdown	+ Log Breakdown → fill form → Save.
Compute a toll	Toll Calculator	Pick expressway → class → entry → exit.
Enter trip toll fee (manual)	Schedule	Type the toll from the physical receipt into the trip's Toll Fee column.
Check today's GPS-detected toll total	Dashboard	Look at the "GPS Toll" KPI card (purple, broadcast icon).
Drill into GPS-detected toll events	Toll Log	Open /toll-log (or click the GPS Toll card on Dashboard).
Export the toll log	Toll Log	Set filters → Export to Excel.
Map a plate to GPS	Master Data	Plates card → broadcast icon → pick vehicle.
Run a poll on demand	Master Data	Plates card toolbar → Poll Now.
Push to Google Sheets	Reports	Set date range → Sync to Sheets.
Pull latest geofences	Truck Cycle Time	Toolbar → Sync Geofences.
See where a truck is right now	Truck Cycle Time	Look at the Plate Status table — open cycles at top.
View a truck's full journey log	Truck Cycle Time	Click the truck's row → pick a cycle chip → audit trail timeline appears below.
Investigate a long cycle	Truck Cycle Time	Filter Cycle Length = Long, then expand the plate row to view the timeline.
Adjust dwell or stop thresholds	Truck Cycle Time	Settings button → modify values → Save (admin only).
Clear trial-period tracking data	Truck Cycle Time	Settings → Danger Zone → pick cutoff date → type CLEAR → confirm (admin only).
Add a manual geofence (workaround)	IT Admin only	Edit manual_geofences.json at project root; polling worker re-reads on next poll.

F.4 Contact Information

Key contacts for system support. Update this table before printing.

Role	Name	Contact
Operations Manager	[TO BE FILLED]	[TO BE FILLED]
Dispatch Supervisor	[TO BE FILLED]	[TO BE FILLED]
Fleet Manager	[TO BE FILLED]	[TO BE FILLED]
Finance Officer	[TO BE FILLED]	[TO BE FILLED]
IT Administrator	[TO BE FILLED]	[TO BE FILLED]
Management Representative	[TO BE FILLED]	[TO BE FILLED]
Cartrack PH Support	Cartrack Customer Care	+63 2 8 555 3000 / support@cartrack.ph

F.5 End of Document

This concludes the Dispatch Scheduler System Manual, version 1.1. The document is controlled by the Management Representative; revisions are logged in the Document Control page (Section A front matter).

BEST PRACTICE **Feedback Welcome**

Errors, omissions, and suggestions for improvement are welcomed. Submit feedback in writing to the Management Representative for inclusion in the next scheduled revision.